



Applying Co-Design/Co-Verification to Multi-Die Designs That Use Advanced Packaging

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January 2025



Agenda & Topics

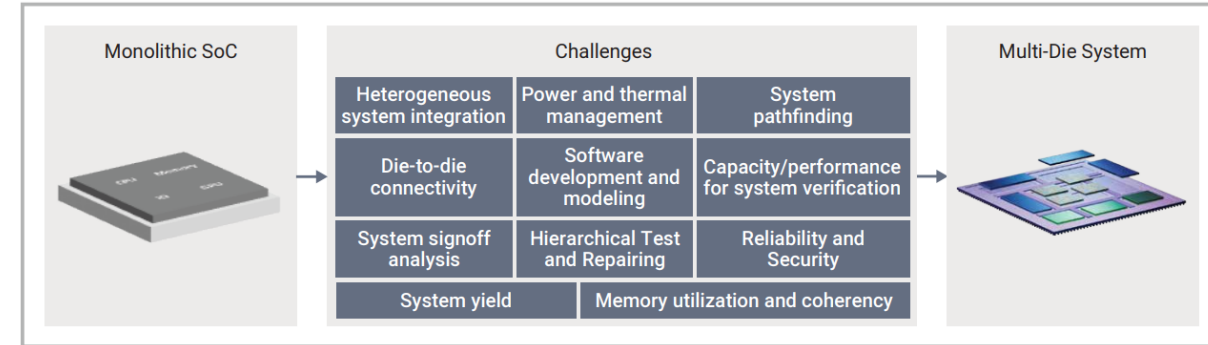
- Advanced Packaging Design-Team Challenges
- Architecture/Partitioning and Layout implementation
- High-Capacity Simulations and sign-off
- Automation to address challenges

Advanced Packaging Design Challenges

Faced by the Design team (there are many other challenges faced by other teams :)

Let's start by establishing that every **Advanced** Package design is different.

Same design goes from CoWoS-S → CoWoS-L → OSAT



Our focus for today

Arch & Design

Design Partitioning

Mega-Design Capacity

Chip-Design flow/tools

Sim & Sign-Off

Chip-Design sim tools new to packaging SI/PI team

Sign-Off process and methodologies

Ecosystem

Establishing partnership with suppliers and foundries

Managing design rules, techfiles and PDKs/ADKs

Certification/Qualification, etc...

Architecture and Layout

SOC Design Partitioning (PPP):

RTL co-optimization + 1.IPs 2.Die Size 3.Floorplan 4.SI/PI 5.Thermal

* New to most packaging teams

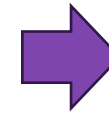
Technology Node Selection ↔ Interface IPs ↔ Die Size ↔ Packaging Technology ↔ Floorplan ↔ Thermal ↔ SIP/PI

Physical Design of Massive Layouts

10s of thousands of nets to route (ie. UCle & HBM)
100s of **power domains** (patterns planning, routing, optimizing)
10s of millions of ubumps, C4s and BGAs
Chip-design platform instead of PCB layout tools

+

Need to be
simulated
co-designed
and
optimized



Engineers with different
skillsets

New Design Methodologies

New Tools (auto routers)

Challenges to Advanced packaging teams:

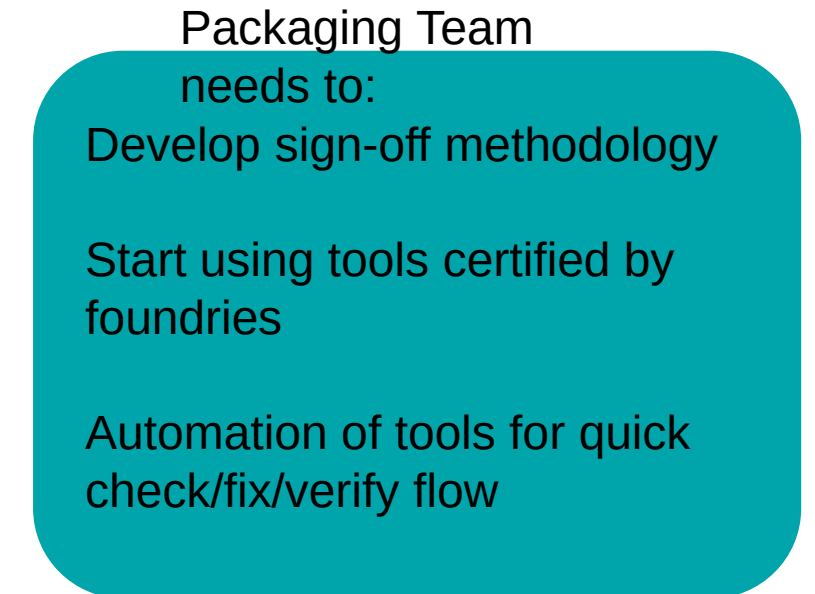
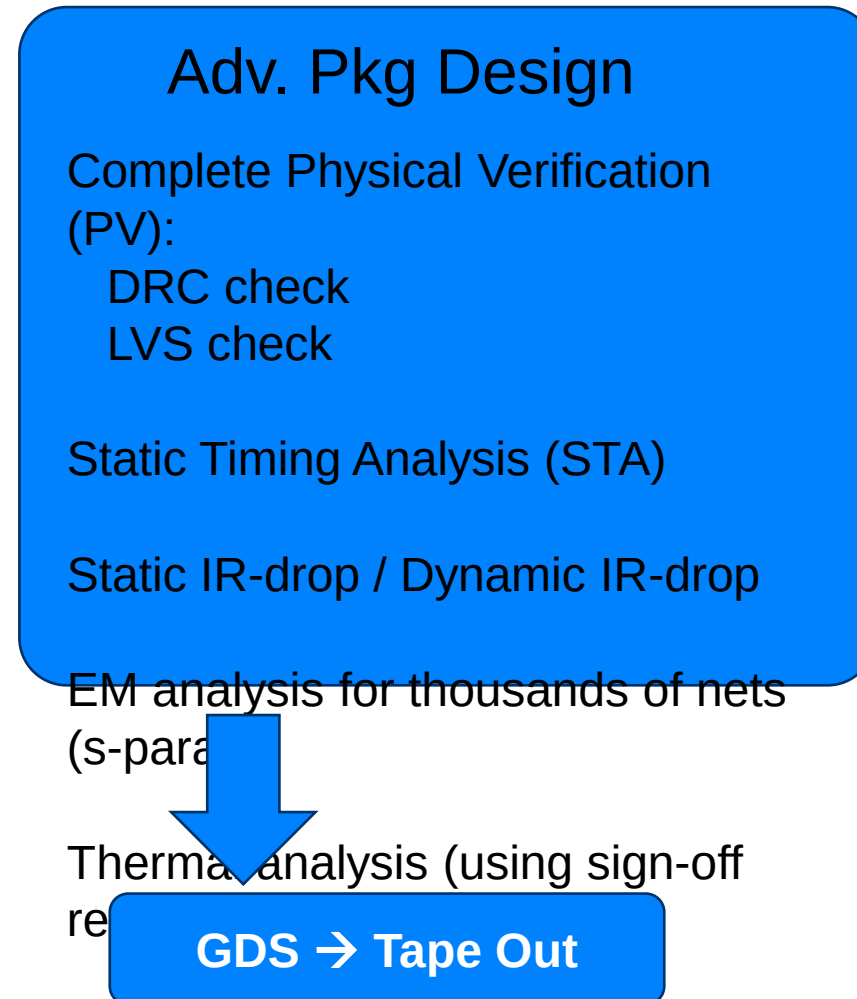
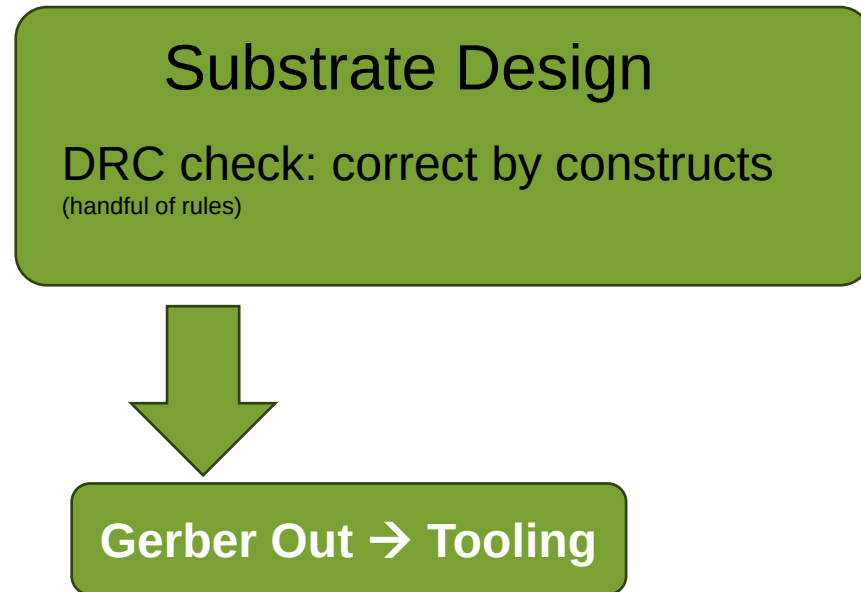
Need to be involved with chip-design process → Rapidly evolving-tools and methodologies

Hopefully, you have not donated your university book for Verilog



Simulation & Sign-Off

Advanced packages, unlike substrates need to go through proper sign-off



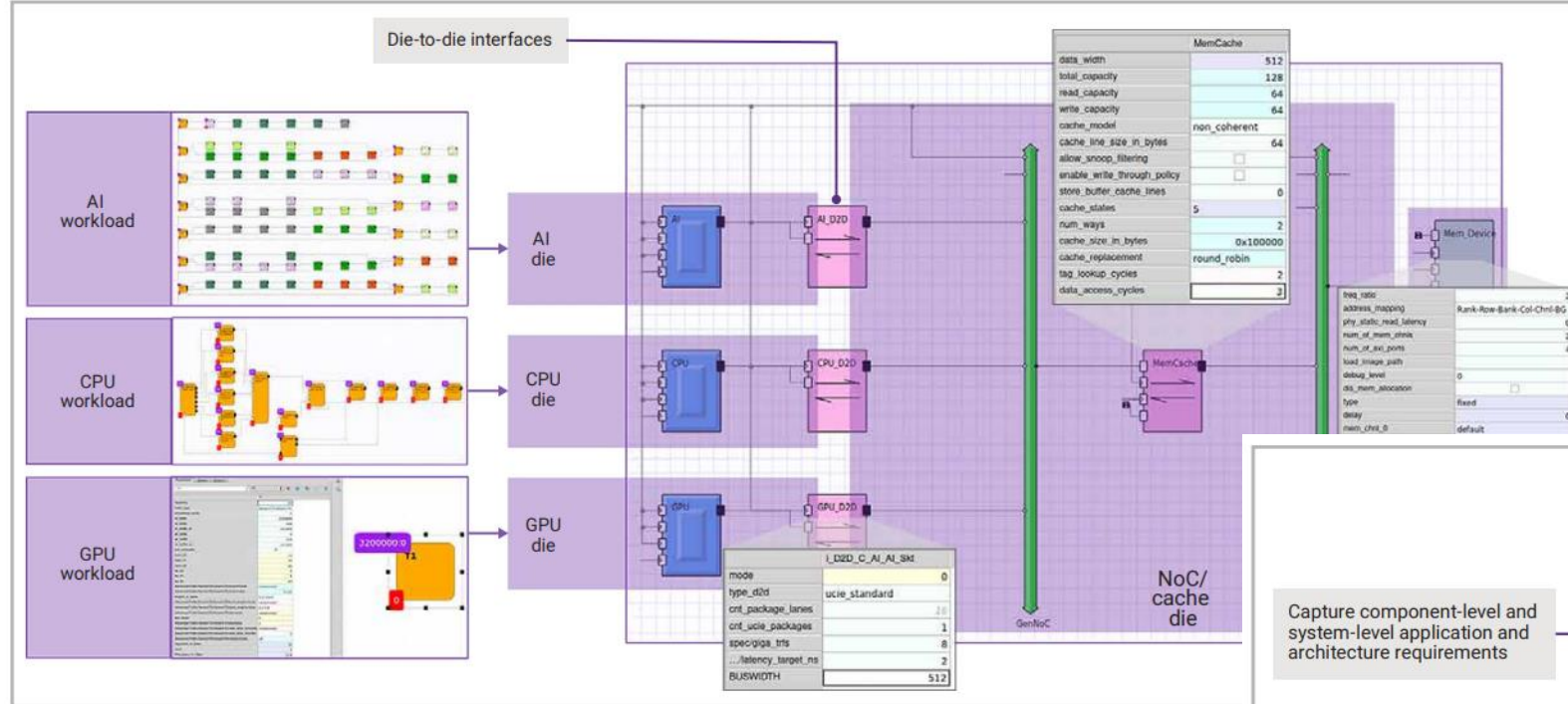
First time I am hiring a dedicated CAD automation engineer for the team!

Overview of key technologies for creating a successful co-design methodology to develop advanced packages

Some of the key tools and platforms used by my team to tackle the challenges

Design Partitioning: RTL and KPIs

Virtual prototyping environment for early architecture analysis/exploration

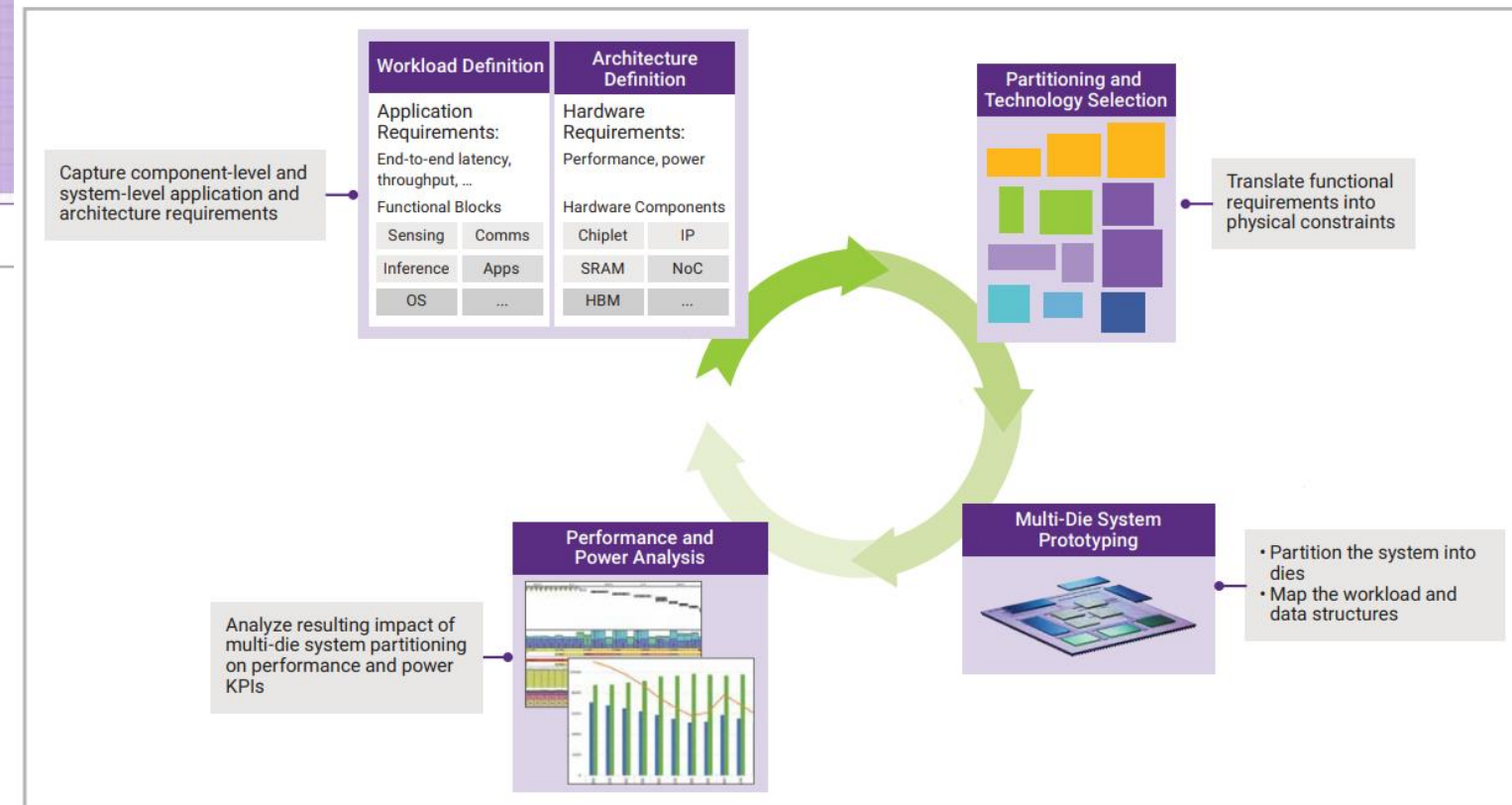


Key Performance Indicators

- Highly configurable models for the system
- Processing and communication requirements of applications as workloads
- System-C, RTL and parametrized models

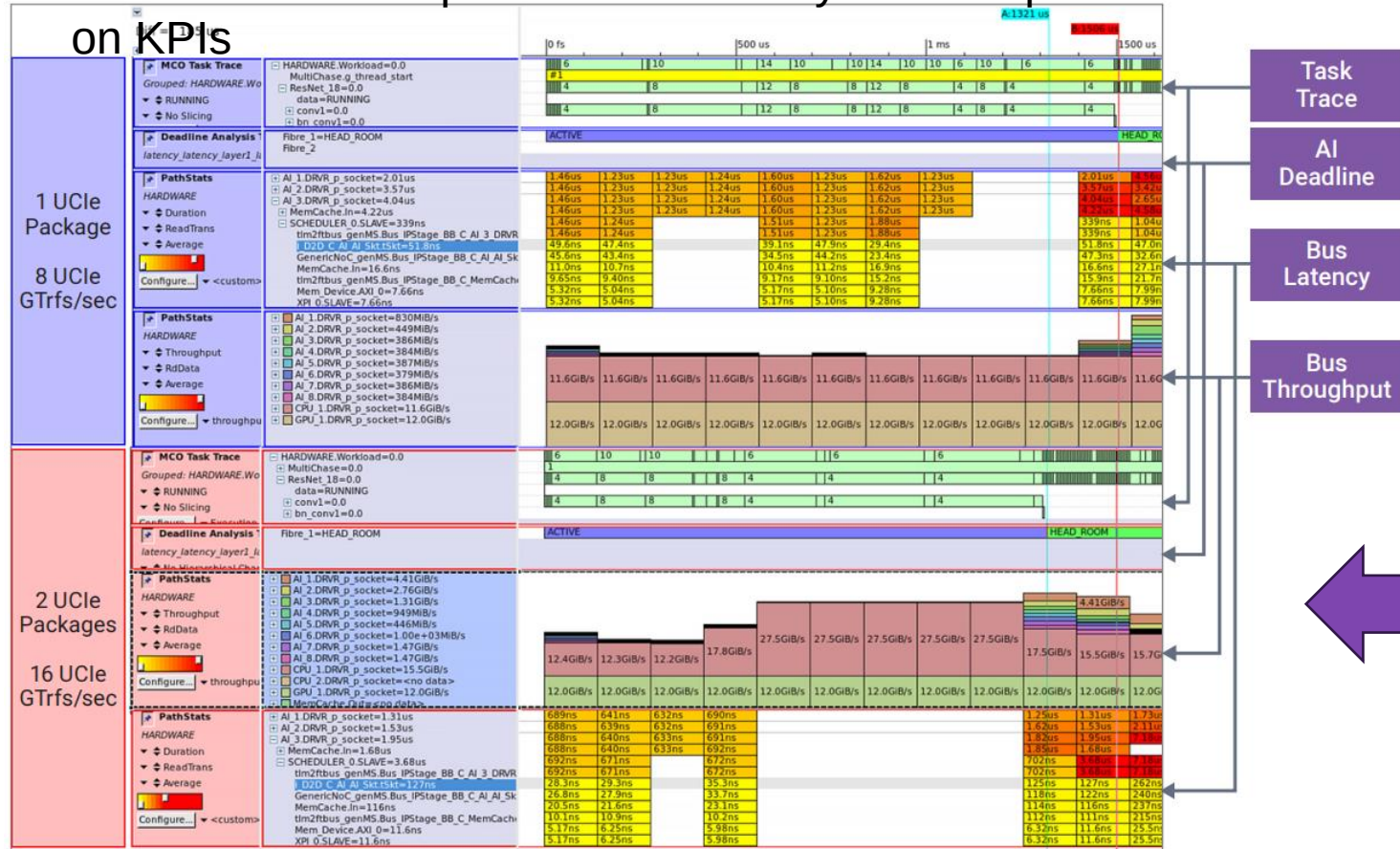
Analyzing the impact of different multi-die system architectures on KPIs

This work is mainly done by the chip-architecture team.
Requires close collaboration with packaging team for key inputs and simulations



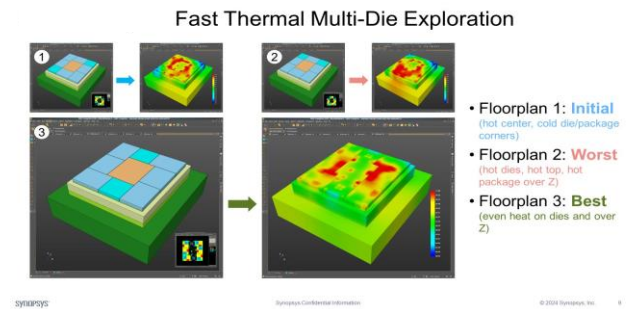
Design Partitioning: Simulation and Performance

Simulate different partitions and analyze the impact on KPIs

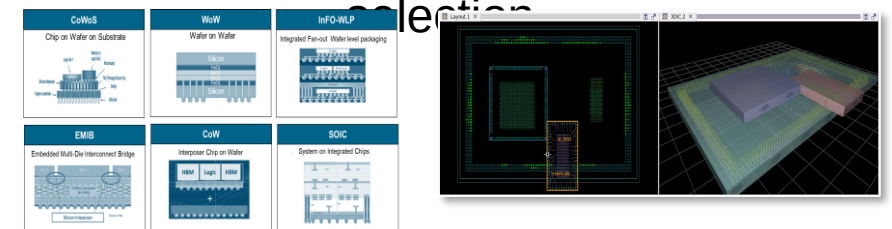


Packaging Team performs analysis on different scenarios

Early Assessment of Thermal/Mechanical performance on different partitions,



Early routability studies and package



Model generation for interfaces and power delivery

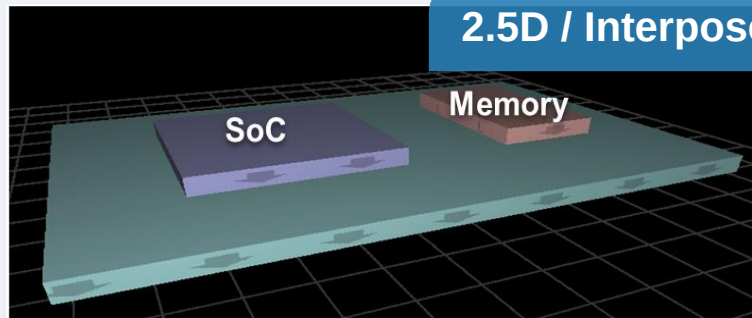


Prototyping Environment for 3DIC Solution Space

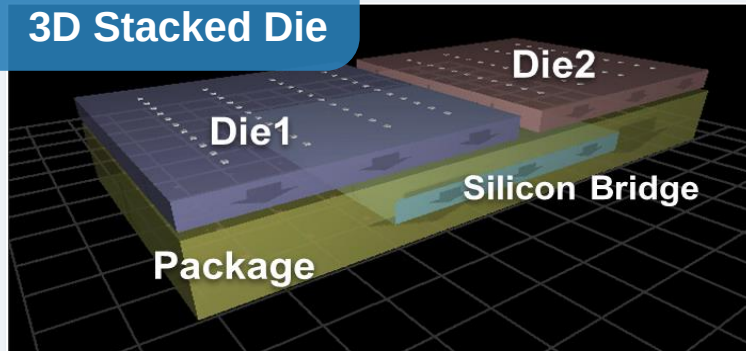
Early specifications-driven, cost-effective selection of optimal configurations

Die abstracts creation

Allows quick placement and thermal and warpage study



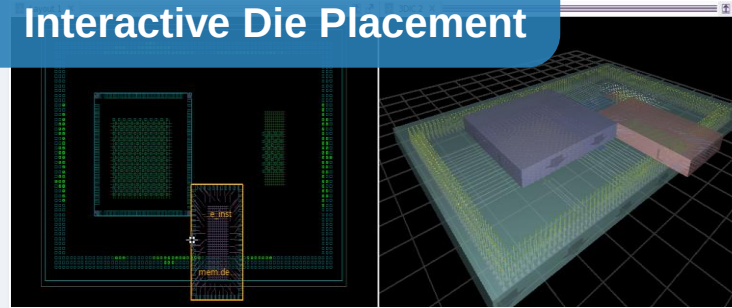
3D Stacked Die



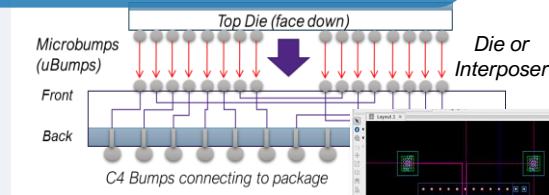
Bump pattern study

PDN bump pattern study, placement, bridge creation & bump alignment...

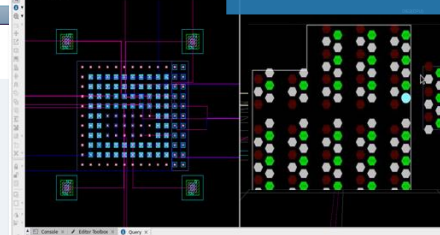
Interactive Die Placement



Auto Bump Mirroring



Interactive Bump Creation



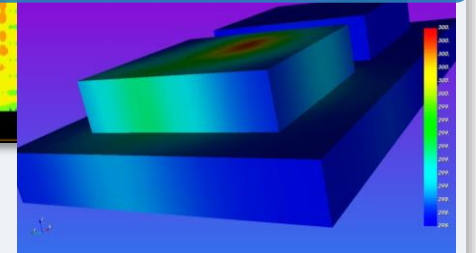
Architecture-level analysis

Golden-Analysis-Driven Optimization

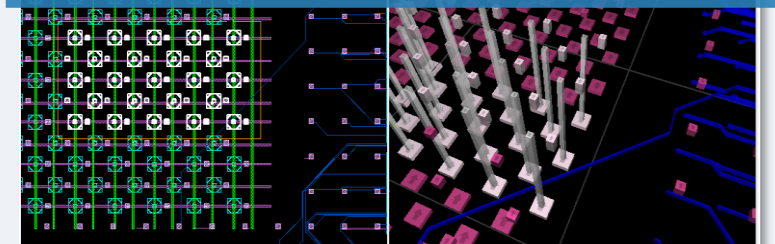
TSV/Power-Driven Optimization



Thermal-Driven Refinement

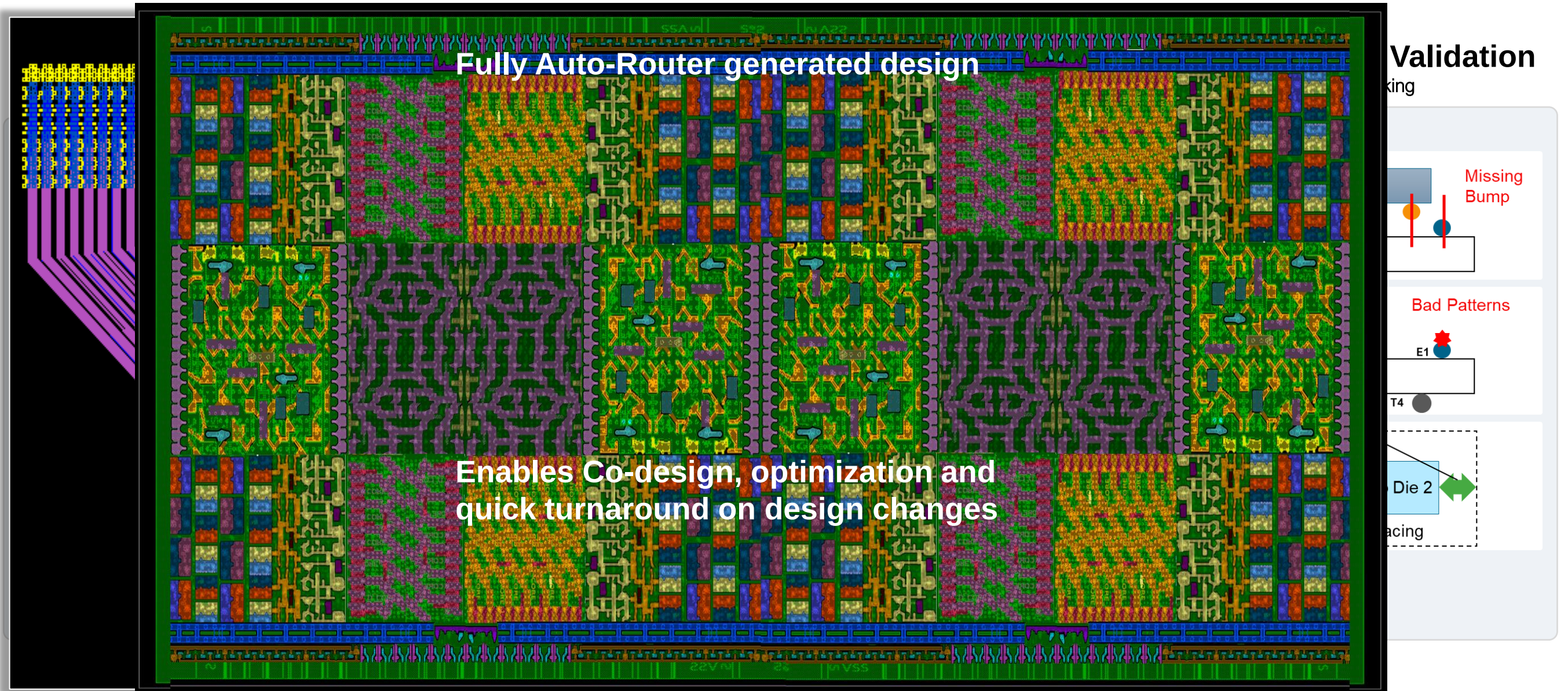


C4 and TSV Selection for Modification



Effective 3DIC Design Implementation

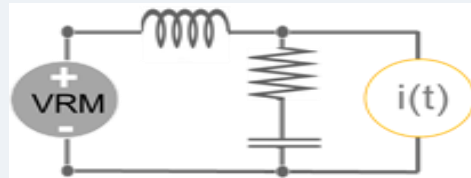
Sign-off ready implementation tool with built-in package router and auto-PDN plane generator



Integrated Multi-Physics Analysis

In-Design multi-die/package analysis and modeling

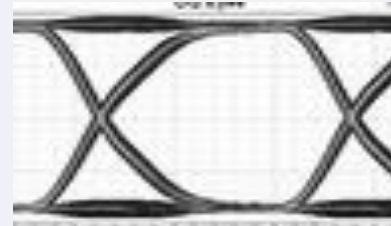
Power Integrity



*Power Delivery Network
vs. Performance*

Physics
Coupling

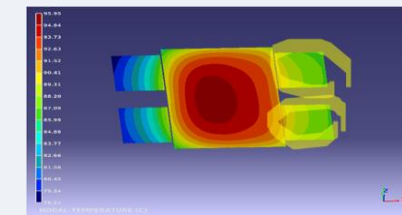
Signal Integrity



*Routing Interconnect
vs. High-Speed Performance*

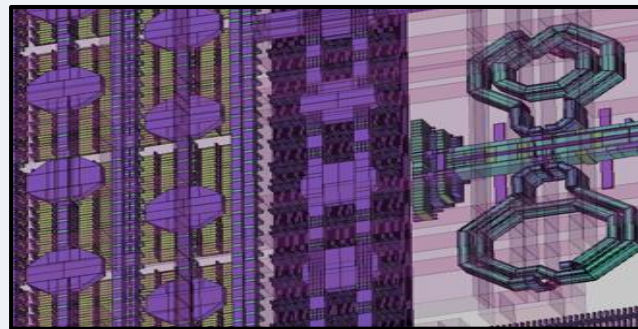
Physics
Coupling

Thermal Integrity

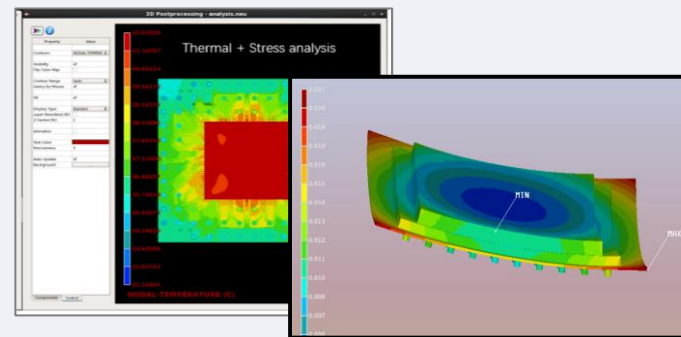


*Power/Heat Dissipation
vs. Reliability*

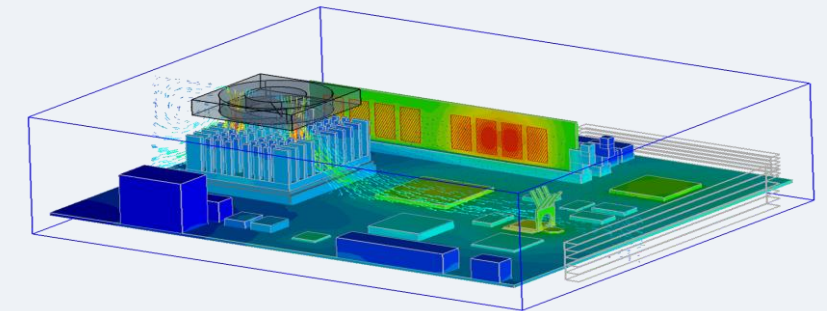
EM Coupling & Interference



Mechanical Stress & Warpage



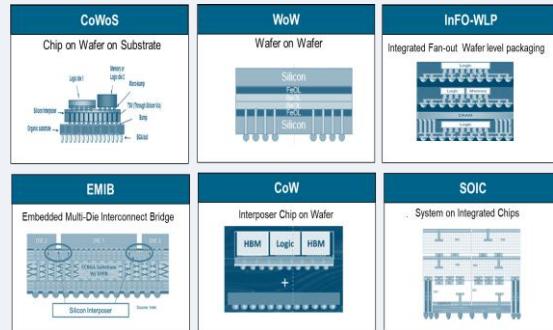
System Thermal



Integrated Full-Spectrum Signoff Analysis

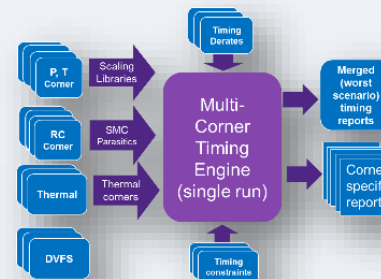
In-Design multi-die analysis delivers productivity coupled with full signoff accuracy

Extraction



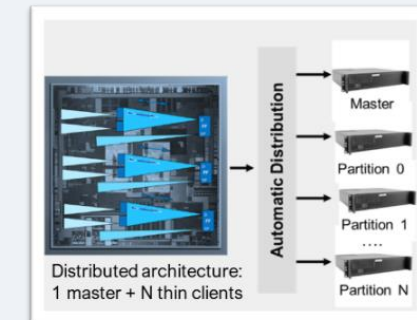
3D Stack/BS, PG and interposer extraction

Static Timing Analysis



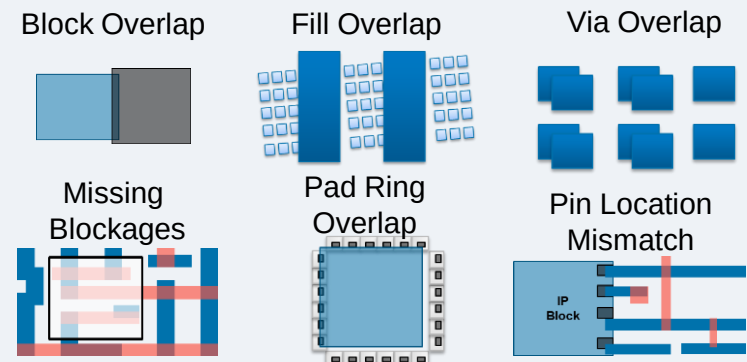
Simultaneous Multi-Corner Analysis

Hypergrid Distributed Analysis

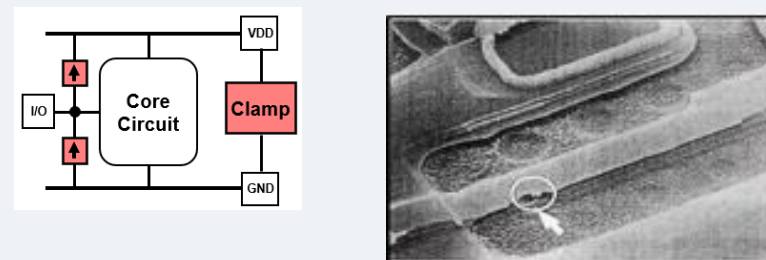


Run 3D STA on billions of instances in hours

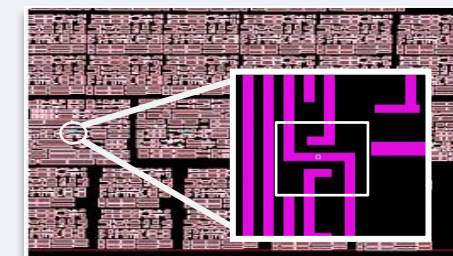
DRC/LVS



ESD

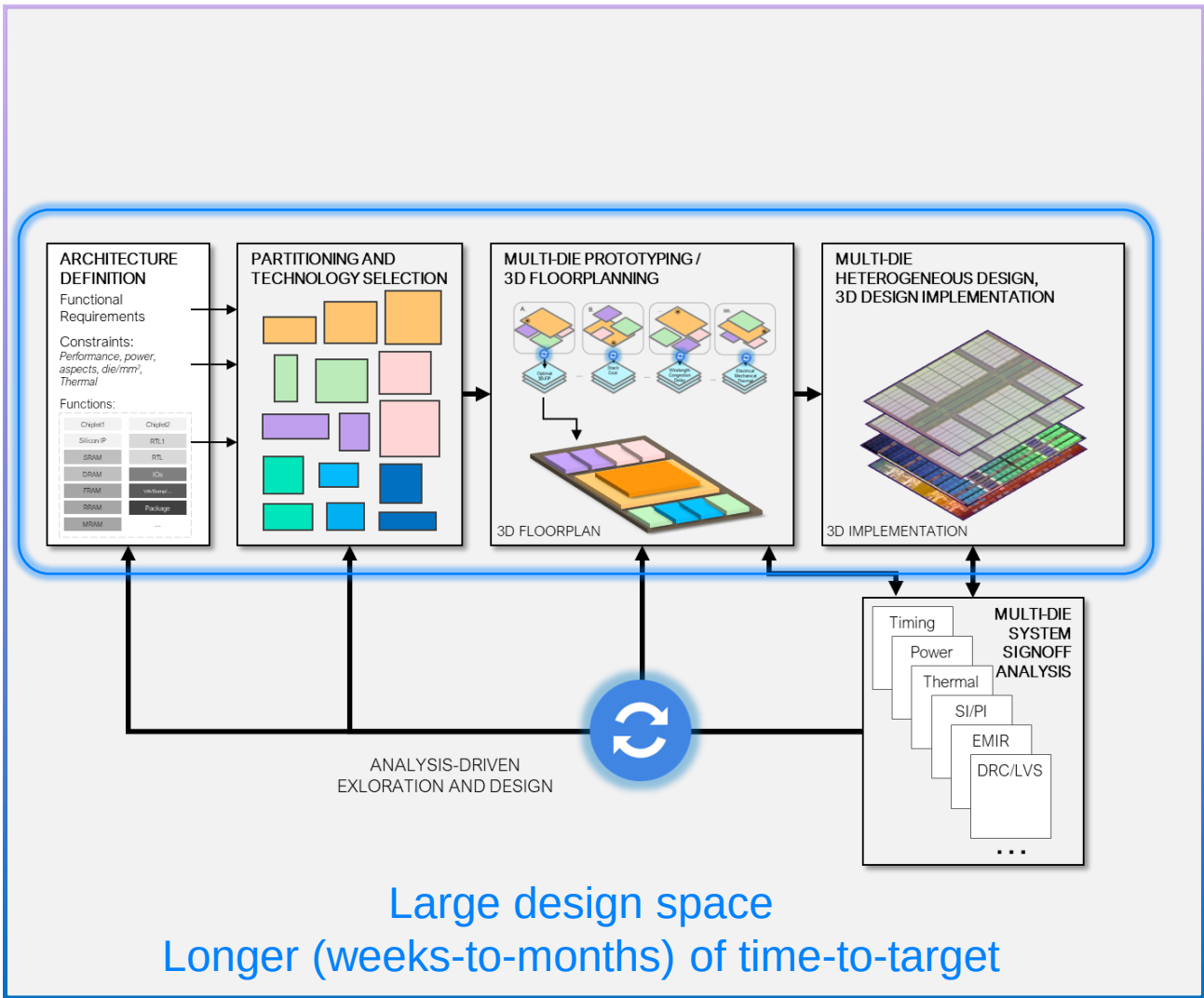


ECO

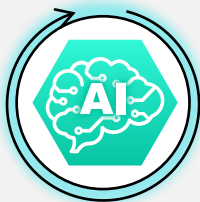


Ultrafast, DRC & Fill ECO within minutes

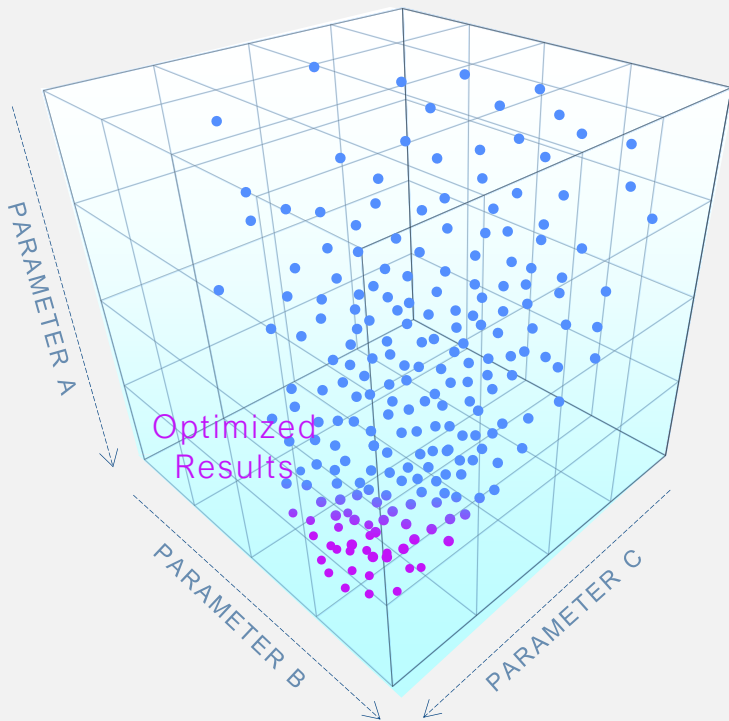
System Analysis: Need for Full Stack Co-Optimization



Opportunity to Accelerate with AI



AI-driven analysis and optimization



Using AI to Improve EDA Tools for Multi-Die Architectures

Synopsys Multi-Die Solution

Comprehensive & Scalable Solution for Fast Heterogeneous Integration



Early Architecture Exploration

Optimize performance, power & thermal with early exploration & partitioning

Software Dev. & Validation

Rapid software development & validation with high-perf. emulation & prototyping

Design Implementation

Efficient die/package co-design & optimization with unified exploration-to-signoff platform

Silicon IP

High-quality, silicon-proven IP portfolio for 2.5D & 3D standard & advanced packaging technologies

Manufacturing & Health

Improve health & reliability from production to in-field with test, analysis & repair solutions

SYNOPSYS®

Thank you